A set is enumerable exactly when it is empty or admits a bijection from the naturals or an initial segment

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openlogic:sets-functions-relations/size-of-sets:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own corollary environment cor:enum-nat-bij (source lines 966-970); the proof environment begins two lines below it and is its own proof, closing at source line 980.

Editorial note. The proof invokes two author results of the same unit by label, namely prop:enum-bij and prop:enum-shift; their statements and their own proofs are retained as context spans placed before the counted statement, so both references resolve inside the assembled package and no source pointer has to be declared.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained items print with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's size-of-sets unit, counted once: enumerability is equivalent to being empty or admitting a bijection from the naturals or a finite initial segment.

Allowed standard prerequisites (editorial).

- Enumerable set. A set is enumerable when it is empty or is the image of a surjection from the positive integers.. It is the starting point of the equivalence chain (source lines 973-974).
- Replacing a surjection by a bijection. A surjection from the positive integers onto A can be converted into a bijection onto A from the positive integers or from a finite initial segment (the author's proposition prop:enum-bij, retained as context).. It is applied in the middle of the chain (source lines 974-977).
- Shifting an enumeration between the positive integers and the naturals. An enumeration indexed by the positive integers can be reindexed by the natural numbers and conversely (the author's proposition prop:enum-shift,

retained as context).. It supplies the last step of the chain, changing the index set from the positive integers to the naturals or an initial segment starting at 0 (source lines 977-980).

Proposition 0.1. *If $f\colon \mathbb{Z}^+ \to A$ is !!surjective (i.e., an enumeration of A), there is !!abijection $g\colon Z \to A$ where Z is either $\mathbb{Z}^+$ or $\{1,\ldots,n\}$ for some $n \in \mathbb{Z}^+$.*

Proof. We define the function g recursively: Let $g(1) = f(1)$. If $g(i)$ has already been defined, let $g(i+1)$ be the first value of $f(1)$, $f(2)$, … not already among $g(1)$, …, $g(i)$, if there is one. If A has just n !!elements, then $g(1)$, …, $g(n)$ are all defined, and so we have defined a function $g\colon \{1,\ldots,n\} \to A$. If A has infinitely many !!elements, then for any i there must be !!aelement of A in the enumeration $f(1)$, $f(2)$, …, which is not already among $g(1)$, …, $g(i)$. In this case we have defined a function $g\colon \mathbb{Z}^+ \to A$.

The function g is !!surjective, since any element of A is among $f(1)$, $f(2)$, … (since f is !!surjective) and so will eventually be a value of $g(i)$ for some i. It is also !!injective, since if there were $j < i$ such that $g(j) = g(i)$, then $g(i)$ would already be among $g(1)$, …, $g(i-1)$, contrary to how we defined g. □

Proposition 0.2. *There is !!asurjection $f\colon \mathbb{Z}^+ \to A$ iff there is !!asurjection $g\colon \mathbb{N} \to A$.*

Proof. Given !!asurjection $f\colon \mathbb{Z}^+ \to A$, we can define $g(n) = f(n+1)$ for all $n \in \mathbb{N}$. It is easy to see that $g\colon \mathbb{N} \to A$ is !!surjective. Conversely, given !!asurjection $g\colon \mathbb{N} \to A$, define $f(n) = g(n-1)$. □

Corollary 0.3. *A set A is !!enumerable iff it is empty or there is !!abijection $f\colon N \to A$ where either $N = \mathbb{N}$ or $N = \{0,\ldots,n\}$ for some $n \in \mathbb{N}$.*

Proof. A is !!enumerable iff A is empty or there is !!asurjective $f\colon \mathbb{Z}^+ \to A$. By 0.1, the latter holds iff there is !!abijective function $f\colon Z \to A$ where $Z = \mathbb{Z}^+$ or $Z = \{1,\ldots,n\}$ for some $n \in \mathbb{Z}^+$. By the same argument as in the proof of 0.2, that in turn is the case iff there is !!abijection $g\colon N \to A$ where either $N = \mathbb{N}$ or $N = \{0,\ldots,n-1\}$. □